Unit 3: Data Warehouse Architecture and OLAP Tools

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System process:

- Before we create architecture for data warehouse, we must first understand the major processes that constitute a data warehouse, some of these processes are shown in the figure below.

System process:

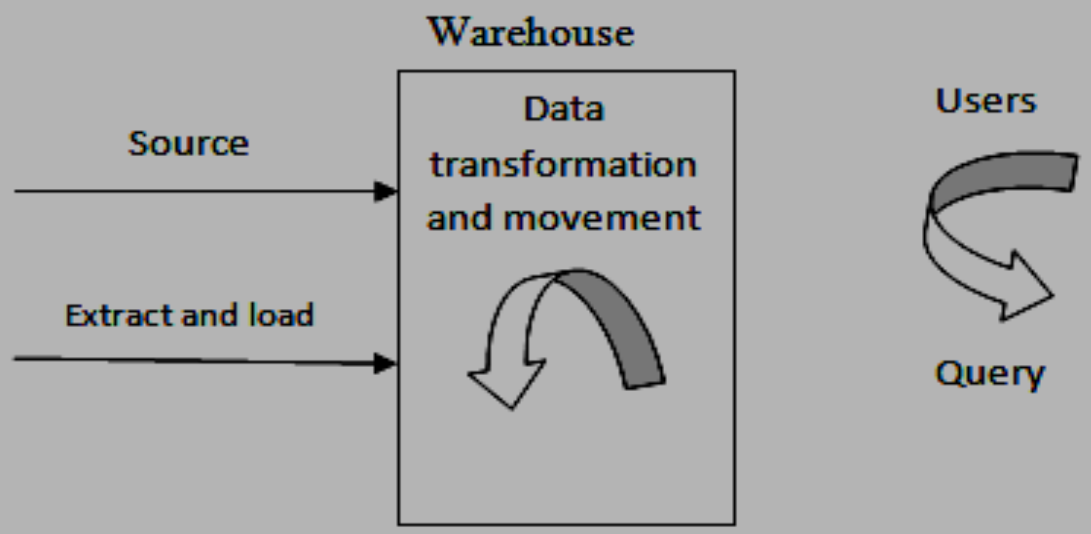


Figure: Process flow within a data warehouse

The processes in above figure are:

- Extract and load data
- Clean and transform data
- Backup and archive data
- Manage queries

1. Extract and load process:

- Data extraction takes data from source systems and makes it available to the data warehouse. Data load takes the extracted data and loads it into the data warehouse.

- When we extract data from a physical database, the original information content will have been modified and extended over the years in order to support the data/performance requirements of the operational system, before loading the data into the data warehouse the information content must be reconstructed.

2. Clean and transform Data:

- This is the system process that takes the loaded data and structures it for query performance and for minimizing operational costs. Some of the steps to be followed are:-
  - Clean and transform the loaded data into a structure that speeds up queries.
  - Partition the data in order to speed up queries, optimize hardware performance and simplify the management of the data warehouse.
  - Create aggregations to speed up the common queries.
- Once the data has been cleaned, next task is to convert the source data into the temporary data storage.

3. Backup and Achieve process:

- The data within the data warehouse is backed up regularly in order to ensure that the data warehouse can always be recovered from data loss, software failure, or hardware failure.

- In archiving, the older data is removed from the system in a format that allows it to be quickly restored if required.

4. Query management process

- It is the system process that manages the queries and speeds that up by directing queries to the most effective data source. The query management process should also monitor the actual query profiles.
- This process operates at all time that the data warehouse is made available to the end users. The query management process determines which table delivers the answer most effectively. It also ensures that so single query can affect the overall system performance.

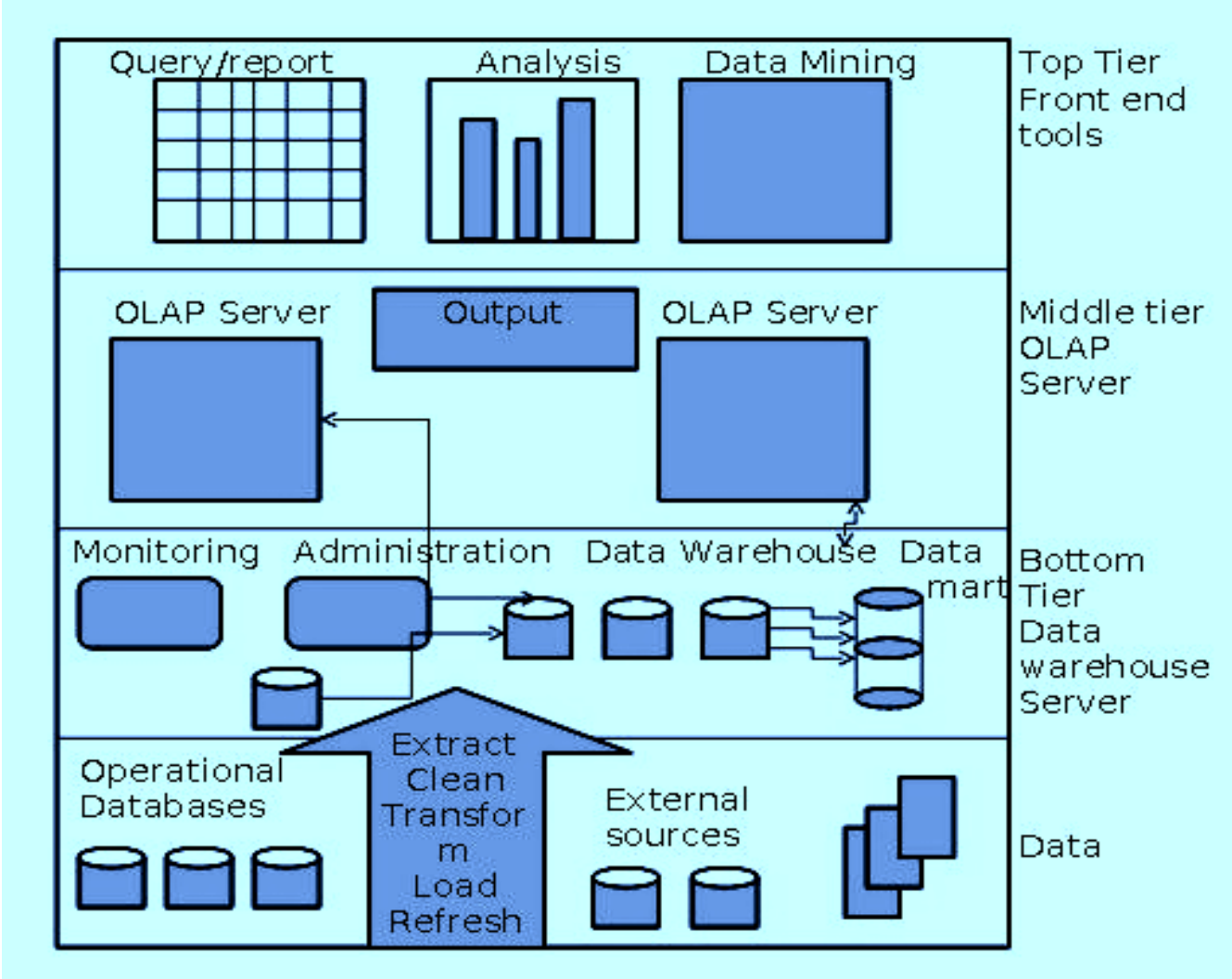
Three-Tier Data Warehouse Architecture

- Generally a data warehouses adopts a three-tier architecture. Following are the three tiers of the data warehouse architecture.
- **Bottom Tier -** The bottom tier of the architecture is the data warehouse database server. It is the relational database system. We use the back end tools and utilities to feed data into the bottom tier. These back end tools and utilities perform the Extract, Clean, Load, and refresh functions.

Three-Tier Data Warehouse Architecture

- **Middle Tier** - In the middle tier, we have the OLAP Server that can be implemented in either of the following ways.
  - By Relational OLAP (ROLAP), which is an extended relational database management system. The ROLAP maps the operations on multidimensional data to standard relational operations.
  -By Multidimensional OLAP (MOLAP) model, which directly implements the multidimensional data and operations.
- **Top-Tier** - This tier is the front-end client layer. This layer holds the query tools and reporting tools, analysis tools and data mining tools.

Three-Tier Data Warehouse Architecture



Data warehouse models

- From the architecture point of view, there are three data warehouse models:
- the enterprise warehouse,
- the data mart, and
- the virtual warehouse.

Enterprise warehouse

- An enterprise warehouse collects all of the information about subjects spanning the entire organization.
- It provides corporate-wide data integration, usually from one or more operational systems or external information providers, and is cross-functional in scope.
- It typically contains detailed data as well as summarized data, and can range in size from a few gigabytes to hundreds of gigabytes, terabytes, or beyond.
- An enterprise data warehouse may be implemented on traditional mainframes, computer super servers, or parallel architecture platforms.
- It requires extensive business modeling and may take years to design and build.

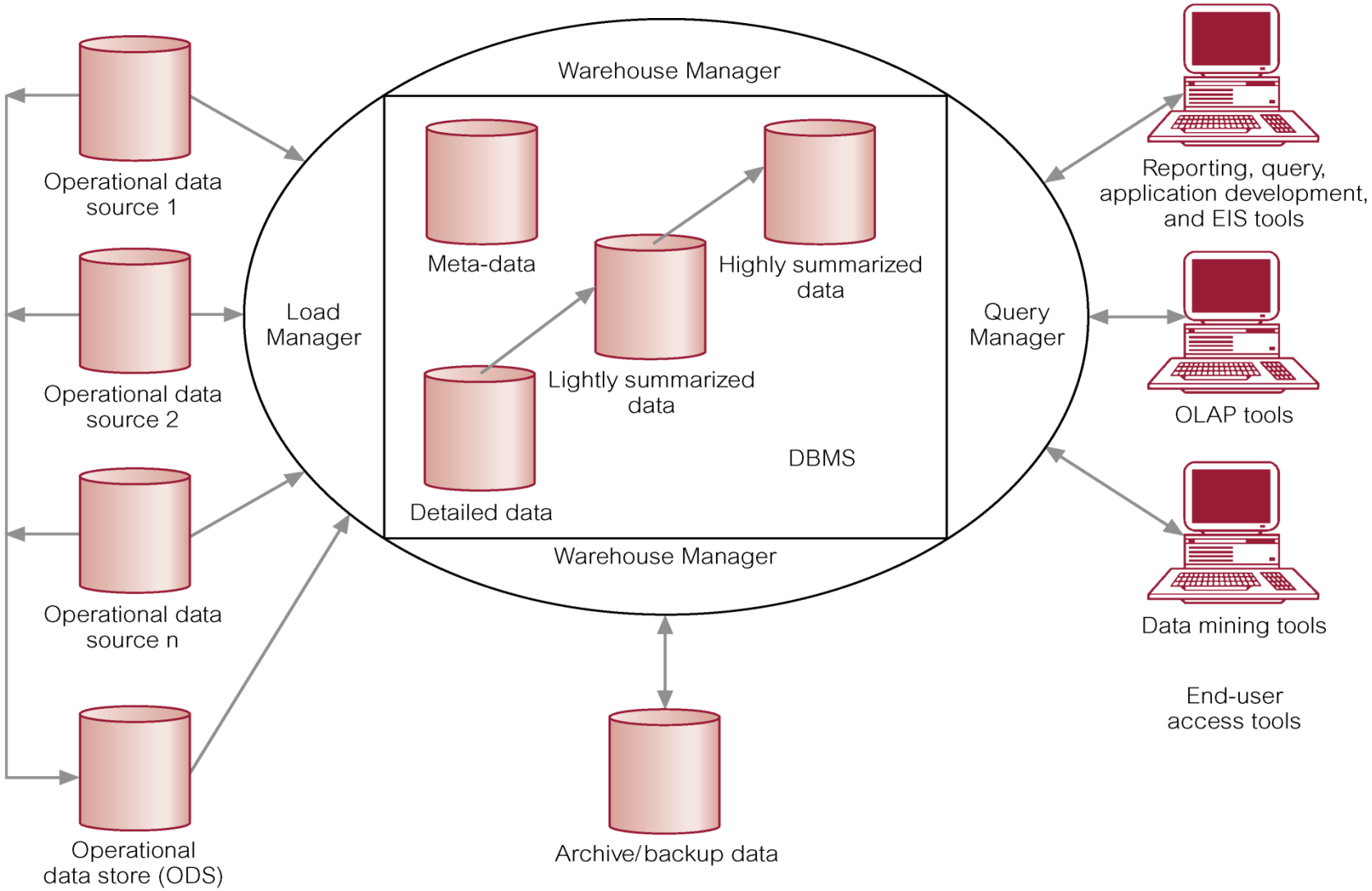
Data mart

- A data mart contains a subset of corporate-wide data that is of value to a specific group of users. The scope is confined to specific selected subjects. For example, a marketing data mart may confine its subjects to customer, item, and sales.
- The data contained in data marts tend to be summarized.
- Data marts are usually implemented on low-cost departmental servers that are UNIX/LINUX- or Windows-based. The implementation cycle of a data mart is more likely to be measured in weeks rather than months or years. However, it may involve complex integration in the long run if its design and planning were not enterprise-wide.
- Depending on the source of data, data marts can be categorized as **independent or dependent**.
- **Independent data marts** are sourced from data captured from one or more operational systems or external information providers, or from data generated locally within a particular department or geographic area.
- **Dependent data marts** are sourced directly from enterprise data warehouses.

Virtual warehouse

- A virtual warehouse is a set of views over operational databases.
- For efficient query processing, only some of the possible summary views may be materialized.
- A virtual warehouse is easy to build but requires excess capacity on operational database servers.

Data Warehouse Architecture



Operational Data Sources: It may include:

- Network databases.
- Departmental file systems and RDBMSs.
- Private workstations and servers.
- External systems (Internet, commercially available databases).

Operational Data Store (ODS): It is a repository of current and integrated operational data used for analysis.

- Often structured and supplied with data in same way as DW.
- May act simply as staging area for data to be moved into the warehouse.
- Provides users with the ease of use of a relational database while remaining distant from decision support functions of the DW.

Warehouse Manager (Data Manager):

- Operations performed include:
  - Analysis of data to ensure consistency.
  - Transformation/merging of source data from temp storage into DW
  - Creation of indexes.
  - Backing-up and archiving data.

Query Manager (Manages User Queries):

- Operations include:
  - directing queries to the appropriate tables and
  - scheduling the execution of queries.
- In some cases, the query manager also generates query profiles to allow the warehouse manager to determine which indexes and aggregations are appropriate.

Meta Data: This area of the DW stores all the meta-data (data about data) definitions used by all the processes in the warehouse.

- Used for a variety of purposes:
  - Extraction and loading processes
  - Warehouse management process
  - Query management process
- End-user access tools use meta-data to understand how to build a query.
- Most vendor tools for copy management and end-user data access use their own versions of meta-data.

Lightly and Highly Summarized Data: It stores all the pre-defined lightly and highly aggregated data generated by the warehouse manager.

- The purpose of summary info is to speed up the performance of queries.
- Removes the requirement to continually perform summary operations (such as sort or group by) in answering user queries.

Archive/Backup Data: It stores detailed and summarized data for the purposes of archiving and backup.

- May be necessary to backup online summary data if this data is kept beyond the retention period for detailed data.
- The data is transferred to storage archives such as magnetic tape or optical disk.

End-User Access Tools:

- The principal purpose of data warehousing is to provide information to business users for strategic decision-making.
- Users interact with the warehouse using end-user access tools.
- There are three main groups of access tools:
  1. Data reporting, query tools
  2. Online analytical processing (OLAP) tools *(Discussed later)*
  3. Data mining tools *(Discussed later)*

Central data warehouse

- Central data warehouse copies and stores all operational and external data and adheres to a single, consistent enterprise data model.
- Central data warehouse may be generated from many individual data marts for better performance and easy accessibility.
- Since central data warehouse only has one data model, the data are consistent and complete to users in this architecture.
- Users only have one environment to logon, without worrying about data in different platforms and environments. And most of the processing is done at the corporate site, therefore it’s very useful or enterprises.

Drawback of Central data warehouse

- It is quite difficult to develop a global data model for most organizations, because finding a single structure for everything is tough.
- It is also difficult to agree on a corporate wide level of detail and naming conventions.
- Organizations must also carefully manage the performance and end-user access to centralized warehouses to ensure that the users continue to rely on the centralized data warehouse.

Distributed data warehouse

- Like distributed file systems and distributed database systems, data warehouses can be distributed in an enterprise.
- These distributed data warehouses (distributed data marts) are consist of many local data warehouses (data marts) and can be accessed by the end users through a "**warehouse front end**".
- This **front end** is a kind of software with a global data image. It also knows the location and format of the needed data and how to send the queries to the final destination. When user asks for something, this software will find the local data warehouse to perform the query and get the result back.

Distributed data warehouse

- Distributed data warehouses may be more flexible, have higher performance and load balancing.
- Besides, it can be designed to match the topology of many organizations, such as a corporate data warehouse at the corporate site and regional data warehouses (the data marts) at the branch offices.
- So corporate queries will go to corporate ones, and local queries will go to regional ones.
- Different technologies can e applied to different data warehouses to better match their subjects.

Drawback of Distributed data warehouse

- Distributed data warehouse still need a global data model, which is, like central data warehouse, hard to develop.
- Besides, global dictionary and topological information make it even more complicated.
- Another disadvantage is performance. If the data warehouses are widely distributed, significant performance degradation and service outages can occur. So it requires considerable trade-offs in data distribution and query optimization.”

Virtual Data Warehouse

- How to build a data warehouse connected with run-time database or legacy systems becomes an interesting topic in the early 1990s, and middleware approach provides an alternative to traditional data warehousing, called virtual data warehouse.
- virtual data warehouses use middleware to build direct connections among disparate applications, whereas traditional data warehouse provides a central repository for enterprise data.
- Middleware is software acting as data hubs and allowing access to the corporate data stored in heterogeneous data sources, like relational databases, legacy systems.

Virtual Data Warehouse

- The most popular approach of virtual data warehouse is “**wrapper**” or “**surround**’ data warehouse, that is the old legacy operational data sources are “wrapped” or “surrounded” by middleware where there are no changes made to the underlying operational systems.

Failure of virtual Data warehouse

- **virtual data warehouse approach failed in several fundamental ways:**

- **Performance:** Operational systems are running for transaction tasks, but OLAP tools may ask for 10,000,000 rows of data in the middle of the on-line transaction processing, and no good database designer wants to design databases for that.
- **Historical data:** Time-variant of data warehouse means a rich amount of historical data, typically 5 to 10 years worth, which provides a foundation for analysis and slicing and dicing data. But operational systems under virtual data warehouse only contain a minimal amount of history – 30 to 60 days typically. As such it cannot organize and manage the historical data because no historical data there.

Failure of virtual Data warehouse

- **Data structure**: The structures in the operational systems are designed to suit the needs of operational processing, without any idea of data analysis, and the "wrapper" software around them can do nothing for optimizing the structure of the data for OLAP.
- **Integration:** The middleware generally will give you access only to data in its raw form. There is no integration in the legacy systems beneath the virtual data warehouse, so OLAP tools that accesses the old legacy data must do the integration by their own.
- **Aggregation:** Data warehouse contains summary data, while operational system does not.

Data Warehouse Manager

- The warehouse manager is the system component that performs all the operations necessary to support the warehouse management process.
- This system is constructed using a combination of 3rd –party system management software, bespose coding, C programs and shell scripts.
- The size and complexity of the warehouse manager will vary between specific solutions.

Architecture Warehouse Manager

❖ The architecture of a warehouse management performs the following operations:

- Analyze the data to perform consistency and referential integrity checks.
- Transform and merge the source data in the temporary data store into the published data.
- Create indexes, business views, partition views, business synonyms against the base data.
- Generate demoralization if appropriate.
- Generate any new aggregations that may be required.
- Update all existing aggregations.
- Backup incrementally or totally the data within the data warehouse.
- Archive the data that has reduced the end of its capture life.

Architecture Warehouse Manager

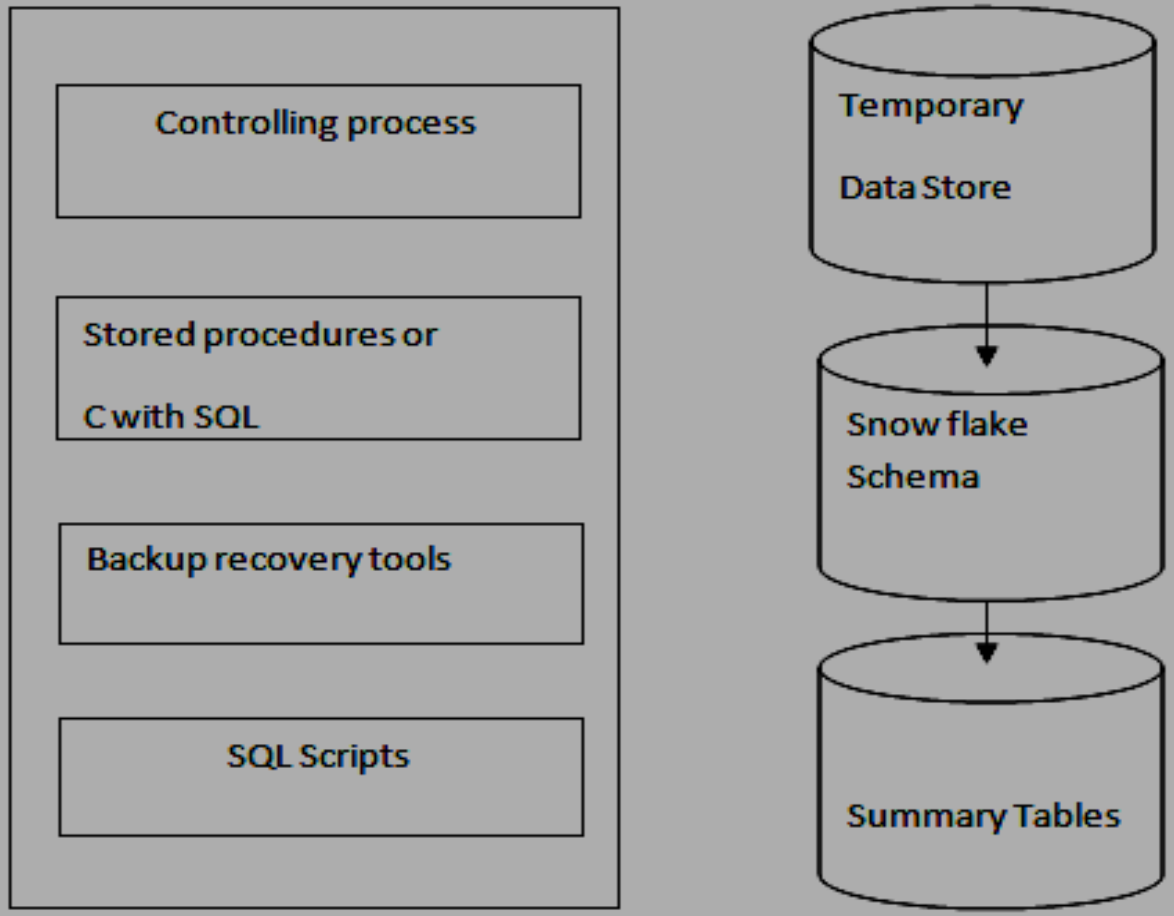


Figure: Architecture of a warehouse Manager

Online analytical processing(OLAP)

- Online analytical processing (OLAP) describes a class of tools that can extract and present multidimensional data from different points of view.
- Designed for managers looking to make sense of their information, OLAP structures data hierarchically – the way managers think of their enterprises.
- OLAP functions include trend analysis, drilling down to more complex levels of detail, summarization of data and data rotation for comparative viewing.

- **OLAP** provides you with a very good view of *what is happening*, but can not predict *what will happen in the future* or *why it is happening*.
- **OLAP** is a term used to describe the analysis of complex data from the data warehouse.
- **OLAP** is an advanced data analysis environment that supports decision making, business modeling, and operations research activities.

- Can easily answer ‘who?’ and ‘what?’ questions, however, ability to answer ‘what if?’ and ‘why?’ type questions distinguishes OLAP from general-purpose query tools.
- Enables users to gain a deeper understanding and knowledge about various aspects of their corporate data through fast, consistent, interactive access to a wide variety of possible views of the data.
- Allows users to view corporate data in such a way that it is a better model of the true dimensionality of the enterprise.

OLAP is a category of applications/technology for

Collecting, managing, processing, and presenting

multidimensional data for analysis and management purposes.

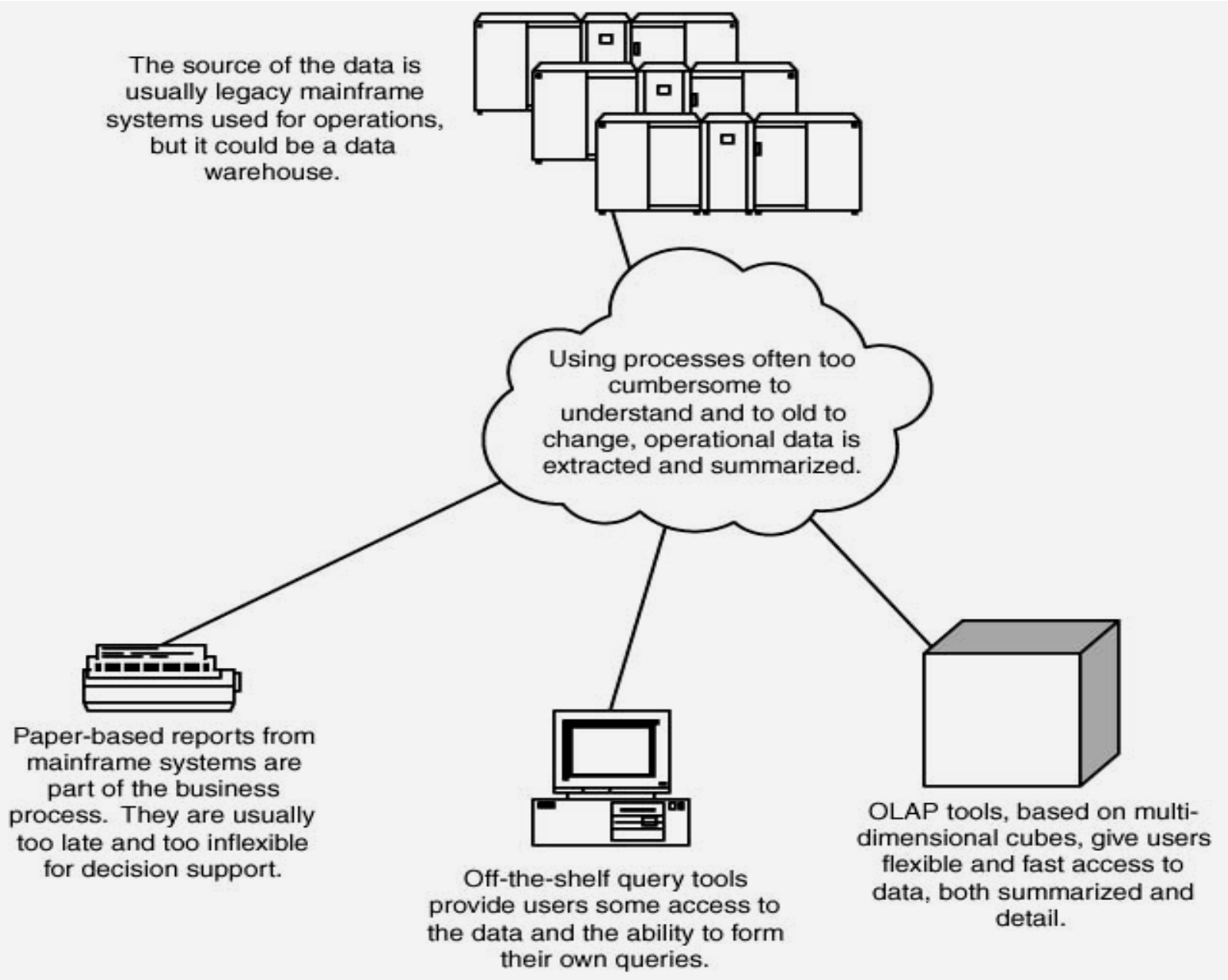
OLAP is **FASMI**

- Fast
- Analysis
- Shared
- Multidimensional
- Information

Comparing OLAP and Data Mining

Characteristic	OLAP	Data Mining
Purpose	Supports data analysis and decision making	Supports data analysis and decision making
Type of analysis supported	Top-down, query-driven data analysis	Bottom-up, discovery-driven data analysis
Skills required of user	Must be very knowledgeable of the data and its business context	Must trust in data-mining tools to uncover valid and worthwhile hypotheses

Where does OLAP fit in?



Examples of OLAP Applications in Various Functional Areas

Functional area	Examples of OLAP applications
Finance	Budgeting, activity-based costing, financial performance analysis, and financial modeling
Sales	Sales analysis and sales forecasting
Marketing	Market research analysis, sales forecasting, promotions analysis, customer analysis, and market/customer segmentation
Manufacturing	Production planning and defect analysis

OLAP Benefits

- Increased productivity of end-users.
- Retention of organizational control over the integrity of corporate data.
- Reduced query drag and network traffic on OLTP systems or on the data warehouse.
- Improved potential revenue and profitability.

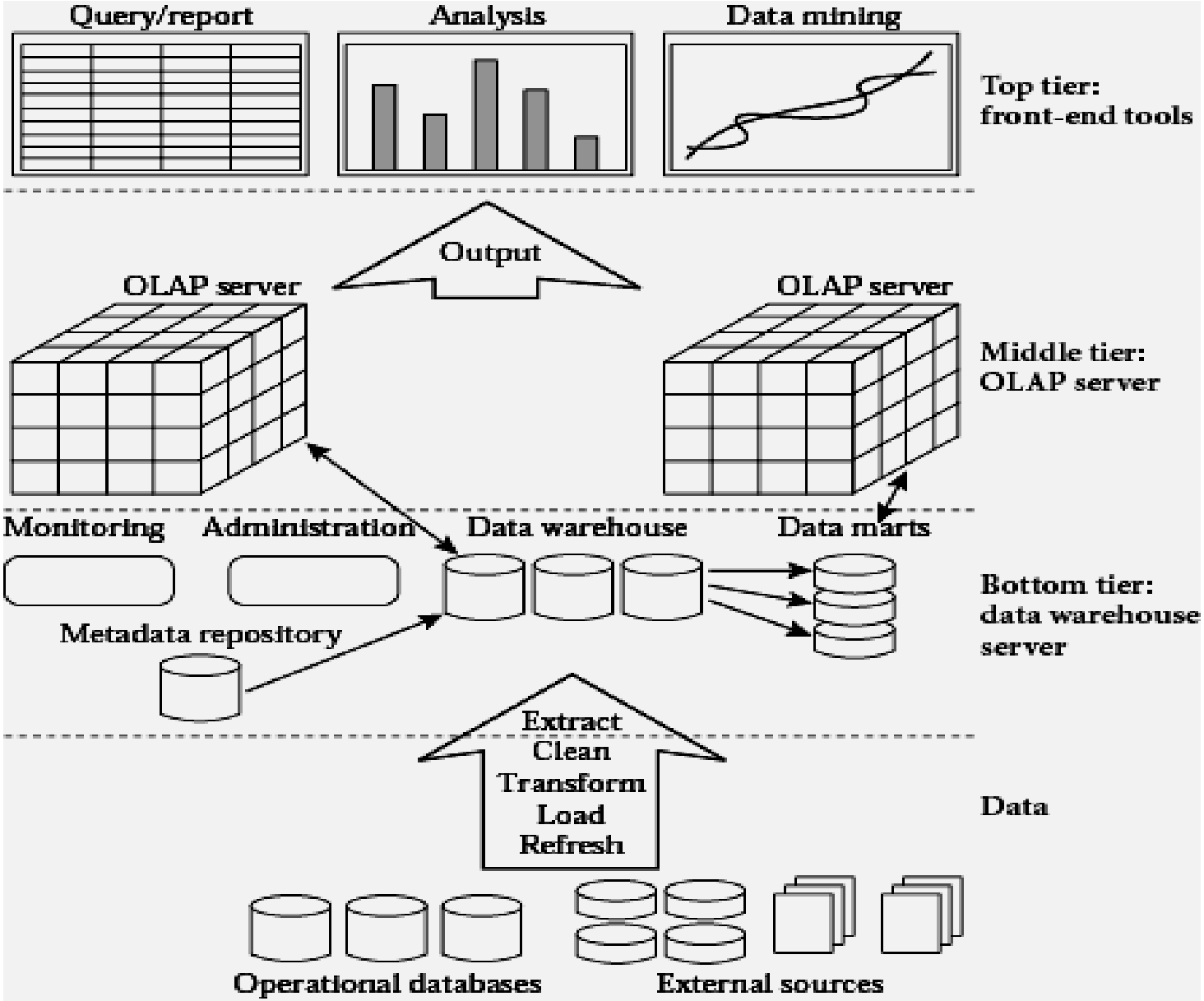
Strengths of OLAP

- It is a powerful visualization paradigm
- It provides fast, interactive response times
- It is good for analyzing time series
- It can be useful to find some clusters and outliers
- Many vendors offer OLAP tools

OLAP for Decision Support

- Goal of OLAP is to support ad-hoc querying for the business analyst
- Business analysts are familiar with spreadsheets
- Extend spreadsheet analysis model to work with warehouse data
  - Large data set
  - Semantically enriched to understand business terms (e.g., time, geography)
  - Combined with reporting features
- Multidimensional view of data is the foundation of OLAP

OLAP Architecture



OLAP Architecture (continued)

- Designed to use both operational and data warehouse data
- Defined as an “advanced data analysis environment that supports decision making, business modeling, and an operation’s research activities”
- In most implementations, data warehouse and OLAP are interrelated and complementary environments

Types of OLAP Server

- OLAP server is classified into four types:

❖Relational OLAP (ROLAP) servers

❖Multidimensional OLAP (MOLAP) servers

❖Hybrid OLAP (HOLAP) servers

❖Specialized SQL Servers

Relational OLAP (ROLAP) servers

- These are the intermediate servers that stand in between a relational back-end server and client front-end tools.
- ROLAP server uses a relational or extended-relational DBMS to store and manage warehouse data, and OLAP middleware to support missing pieces.
- ROLAP servers include optimization for each DBMS back end, implementation of aggregation navigation logic, and additional tools and services.
- ROLAP technology tends to have greater scalability than MOLAP technology.

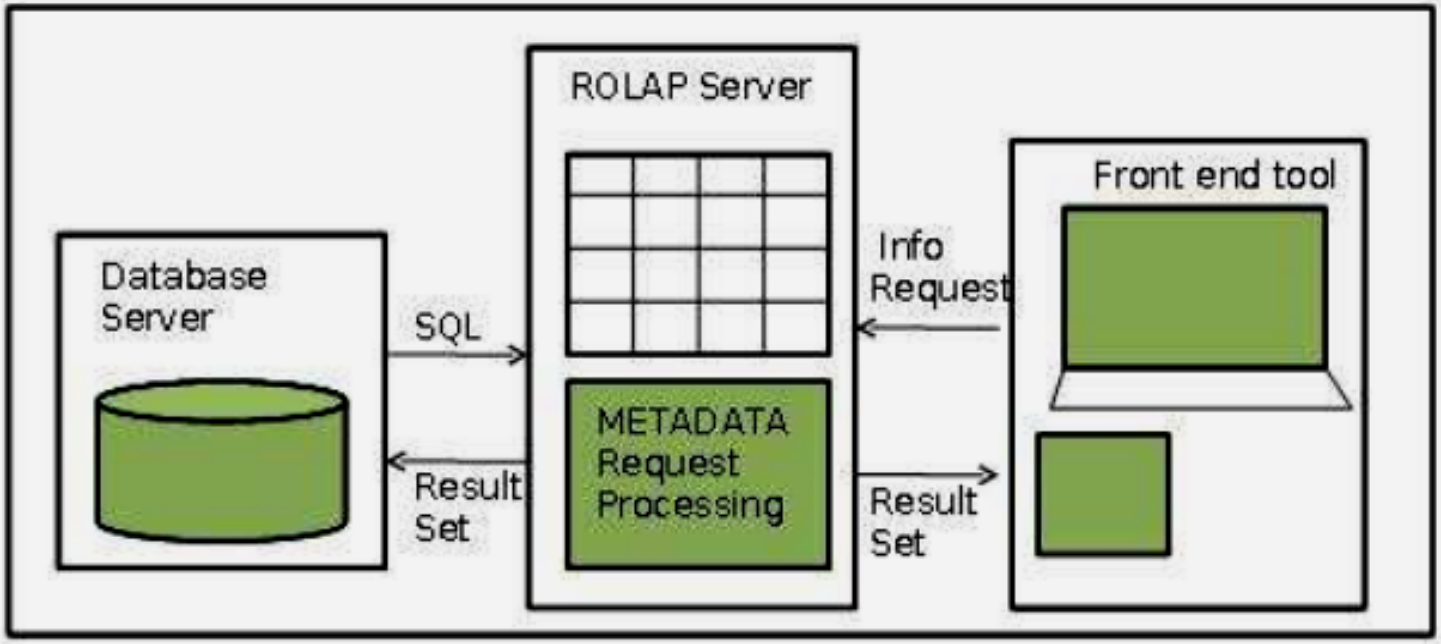
Relational OLAP Architecture

ROLAP includes the following components:

Database server

ROLAP server

Front-end tool.



Advantage of ROLAP

- ROLAP servers can be easily used with existing RDBMS.
- Data can be stored efficiently, since no zero facts can be stored.
- ROLAP tools do not use pre-calculated data cubes.
- DSS server of micro-strategy adopts the ROLAP approach.

Disadvantage of ROLAP

- Poor query performance.
- Some limitations of scalability depending on the technology architecture that is utilized.

Multidimensional OLAP (MOLAP) servers

- These servers support multidimensional views of data through array-based multidimensional storage engines.
- MOLAP servers map multidimensional views directly to data cube array structures.
- The advantage of using a data cube is that it allows fast indexing to pre computed summarized data.
- With multidimensional data stores, the storage utilization may be low if the data set is sparse.
- MOLAP servers adopt a two-level storage representation to handle dense and sparse data sets: **denser sub cubes** are identified and stored as array structures, whereas **sparse sub cubes** employ compression technology for efficient storage utilization.
- MOLAP **main advantage** is it has high processing speed (i.e. faster computation).
- Its main **drawback** is it needs large storage requirements.

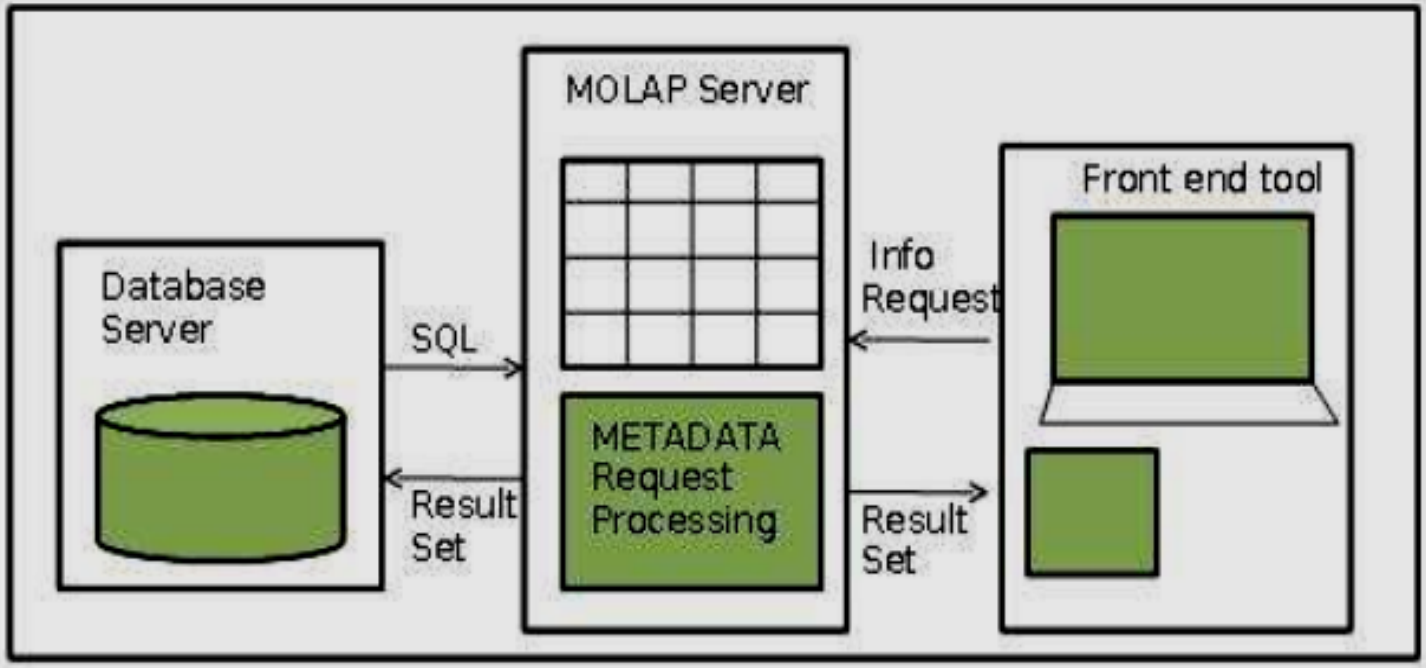
MOLAP Architecture

MOLAP includes the following components:

Database server.

MOLAP server.

Front-end tool.



Advantage of MOLAP

- MOLAP allows fastest indexing to the pre-computed summarized data.
- Helps the users connected to a network who need to analyze larger, less-defined data.
- Easier to use, therefore MOLAP is suitable for inexperienced users.

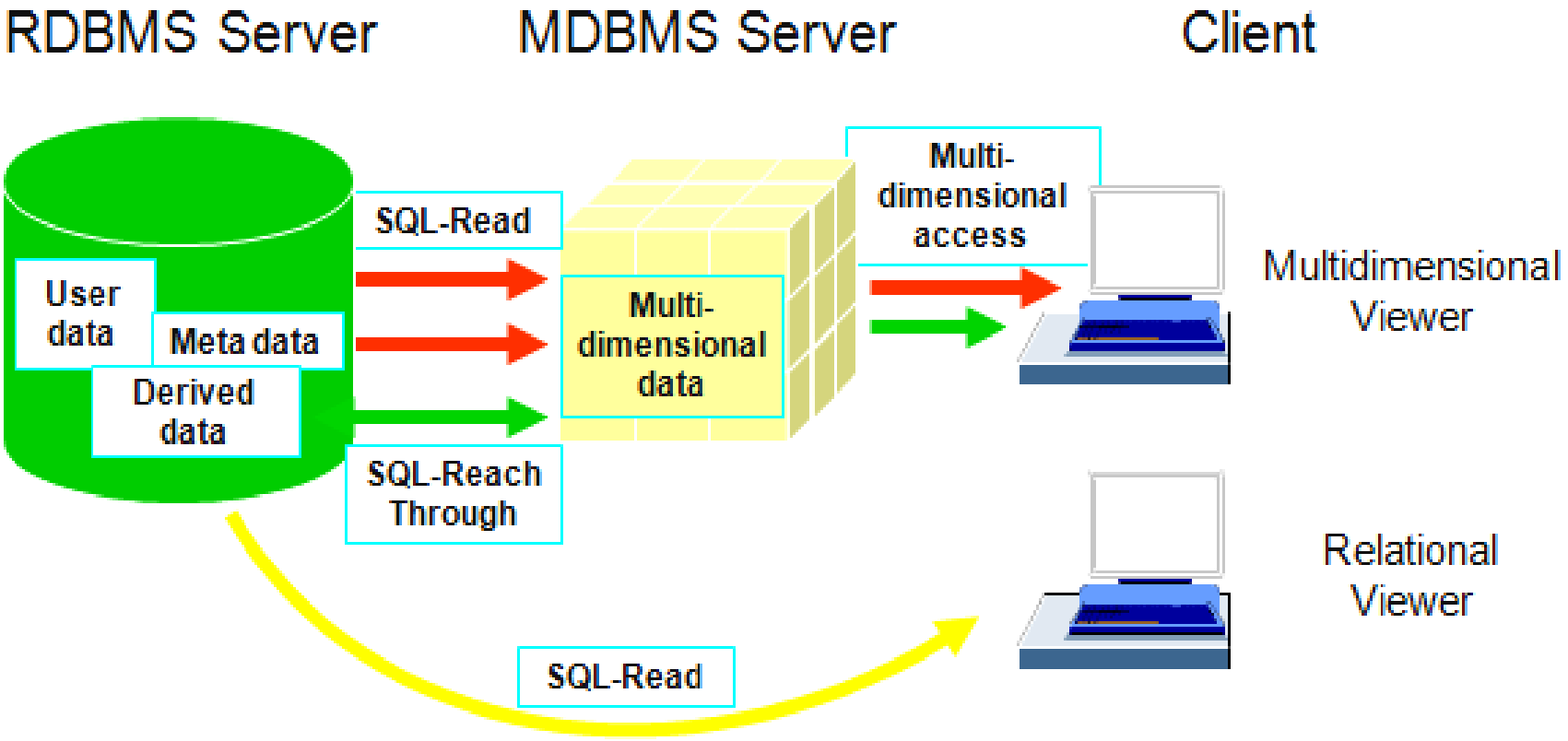
Disadvantage of MOLAP

- MOLAP are not capable of containing detailed data.
- The storage utilization may be low if the data set is sparse.

Hybrid OLAP (HOLAP) servers

- Hybrid OLAP combines the best features of ROLAP and MOLAP
- Data that are not updated frequently will be stored in multidimensional databases whereas data that are updated regularly will be kept in relational databases.
- HOLAP has the benefits of high scalability features of ROLAP and high computing speed of MOLAP.

Data Flow in HOLAP



HOLAP Features:

- For summary type info – cube, (Faster response)
- Ability to drill down – relational data sources (drill through detail to underlying data)
- Source of data transparent to end-user

Specialized SQL Servers

- Specialized SQL servers provide advanced query language and query processing support for SQL queries over star and snowflake schemas in a read-only environment.

Thank you !!!